



MONASH University

MONASH UNIVERSITY

Faculty of Information Technology

FIT5032 Internet Applications Development

Assessed Lab 10

API Service Integration

Weather and Library JSON APIs

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Tutorial/Lab	Tuesday 2:00 pm
Project Name	hwang-library
GitHub Repository	https://github.com/kokyuu/hwang-library

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1. Overview

This report documents the completed FIT5032 Assessed Lab 10 implementation in the existing `hwang-library` Vue 3 project. The application integrates an external weather service, supports browser-location and city-based weather requests, calculates author and book statistics from the Lab 2 dataset, and exposes all books as a JSON-formatted API page.

Assessment item	Implemented evidence
Task 10.1 - weather	Current-location method and weather result UI
Task 10.1 - statistics	3 authors and 6 books in <code>CountBookAPI</code>
Task 10.2 - city search	Clayton, AU result in Celsius with a weather icon
Task 10.2 - all books	6 books shown as JSON on <code>GetAllBookAPI</code>
Version control	Dedicated <code>lab10-api</code> branch and focused commits

2. Page Setup and Routing

The application uses Vue Router and a shared `BHeader` navigation component. The route name `GetWeather` maps to `/WeatherCheck`, while the two library response pages are available at `/CountBookAPI` and `/GetAllBookAPI`.

Lab 10 routes

`src/router/index.ts`

```

1  import { createRouter, createWebHistory } from 'vue-router'
2  import CountBookAPI from '../views/CountBookAPI.vue'
3  import GetAllBookAPI from '../views/GetAllBookAPI.vue'
4  import WeatherView from '../views/WeatherView.vue'
5
6  const router = createRouter({
7    history: createWebHistory(import.meta.env.BASE_URL),
8    routes: [
9      { path: '/', redirect: '/WeatherCheck' },
10     { path: '/WeatherCheck', name: 'GetWeather', component: WeatherView },
11     { path: '/CountBookAPI', name: 'CountBookAPI', component: CountBookAPI },
12     { path: '/GetAllBookAPI', name: 'GetAllBookAPI', component: GetAllBookAPI },
13   ],
14 })
15
16 export default router

```

Figure 1. Vue Router configuration for all Lab 10 pages.

3. Task 10.1 - Weather API

3.1 Current location and city controls

The WeatherView component validates a city search, handles loading and error states, and uses the browser Geolocation API for the current-location workflow. Coordinates are sent only after the user selects the Use current location control.

City search and browser geolocation

src/views/WeatherView.vue

```
38  async function searchByCity() {
39    const query = city.value.trim()
40    if (!query) {
41      error.value = 'Enter a city name to search.'
42      return
43    }
44
45    await loadWeather(() => searchWeatherByCity(query))
46  }
47
48  async function fetchCurrentLocationWeather() {
49    if (!navigator.geolocation) {
50      error.value = 'Geolocation is not supported by this browser.'
51      return
52    }
53
54    loading.value = true
55    error.value = ''
56    navigator.geolocation.getCurrentPosition(
57      async ({ coords }) => {
58        await loadWeather(() => getWeatherByCoordinates(coords.latitude, coords.longitude))
59      },
60      (geolocationError) => {
61        loading.value = false
62        error.value = `Unable to use current location: ${geolocationError.message}`
63      },
64      { enableHighAccuracy: false, timeout: 30000, maximumAge: 300000 },
65    )
66  }
67
68  async function loadWeather(loader: () => Promise<WeatherResult>) {
69    loading.value = true
70    error.value = ''
71    try {
72      weatherData.value = await loader()
73    } catch (requestError) {
74      const message = requestError instanceof Error ? requestError.message : 'Weather request failed.'
75      error.value = message
76    } finally {
77      loading.value = false
78    }
79  }
```

Figure 2. City-search and browser-geolocation methods in WeatherView.vue.

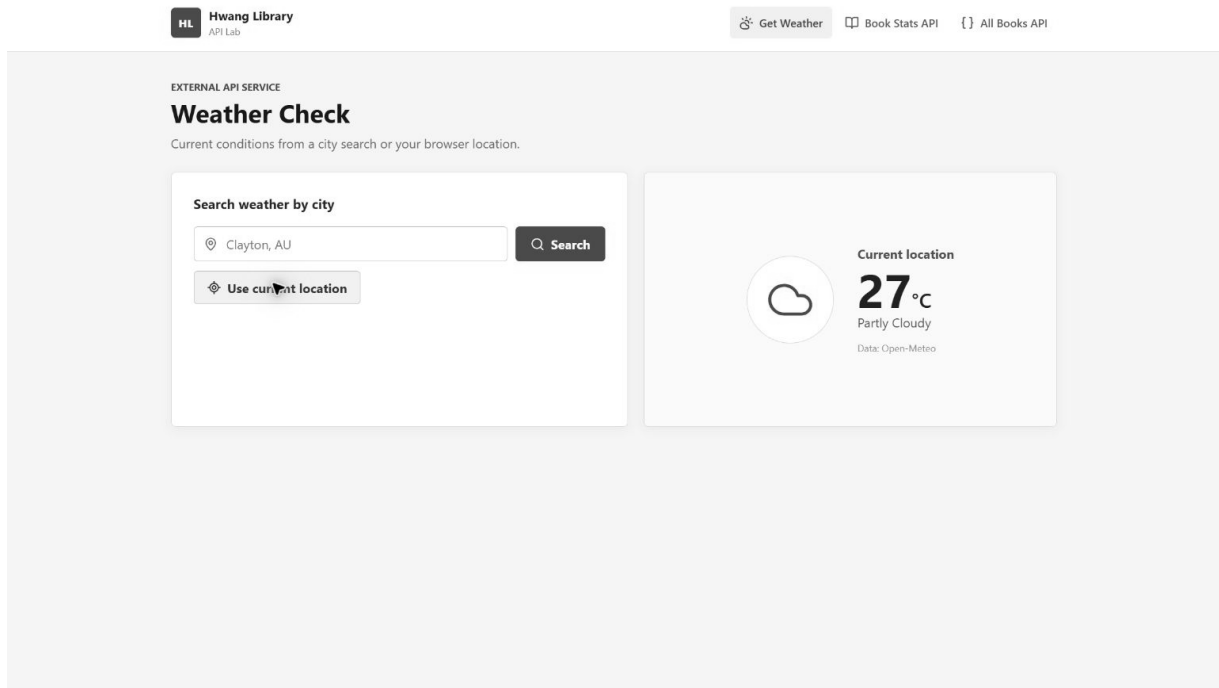


Figure 3. Browser result for the authorised current location in Celsius using Open-Meteo.

3.2 Search weather by city

The external service accepts the assessed-lab example Clayton, AU. Country-code matching prevents the request from selecting a same-named city in another country. The returned Celsius temperature and weather code drive the corresponding weather icon.



Figure 4. Weather API request, country matching and result mapping.

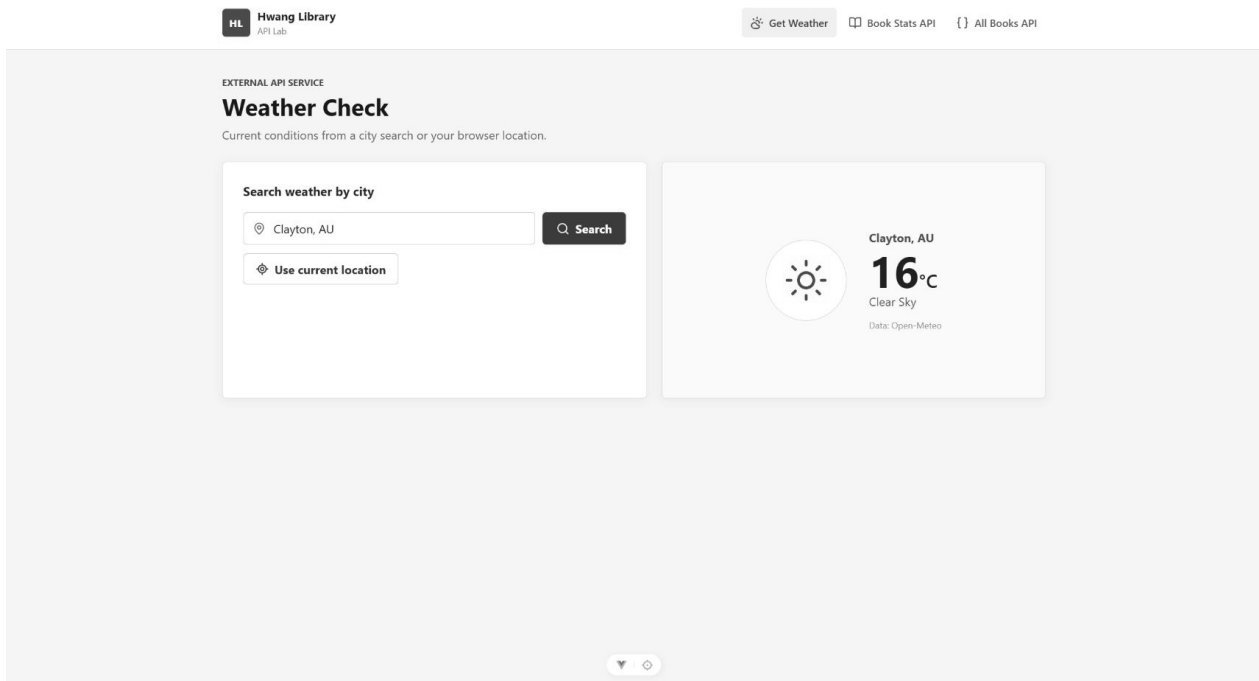


Figure 5. Browser result for Clayton, AU in Celsius with a clear-sky icon.

4. Task 10.1 - Author and Book Statistics

CountBookAPI imports the Lab 2 authors.json file, calculates the number of authors, reduces the nested famousWorks arrays into a total book count, and returns per-author counts. The response correctly reports 3 authors and 6 books.

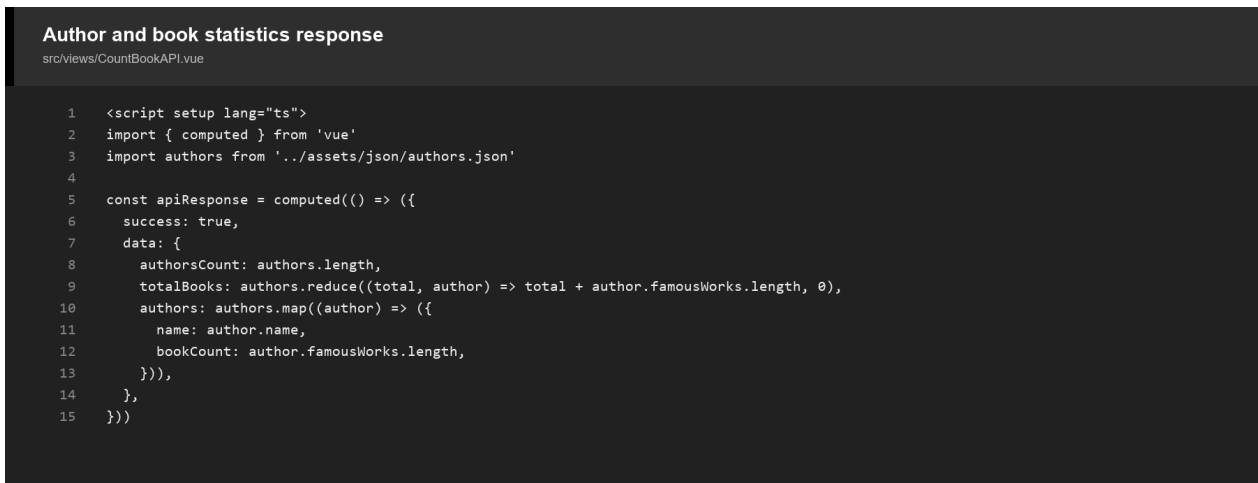


Figure 6. CountBookAPI response calculation from authors.json.

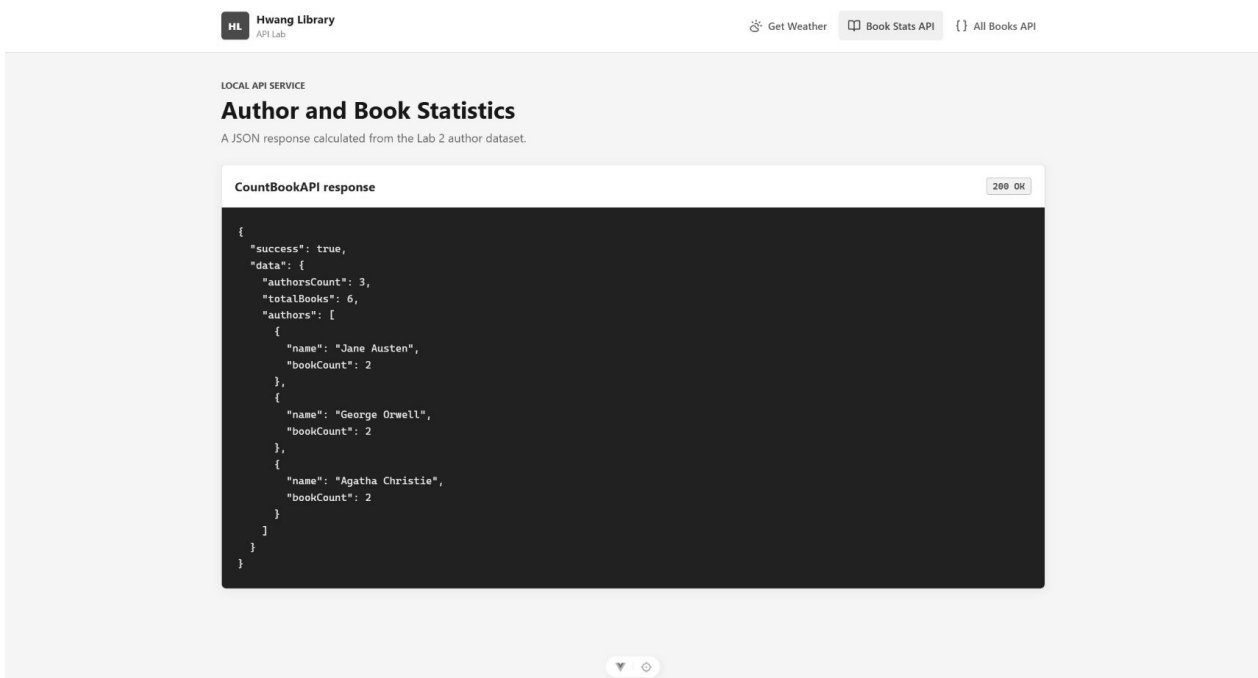


Figure 7. Browser response showing authorsCount 3 and totalBooks 6.

5. Task 10.2 - GetAllBookAPI

GetAllBookAPI flattens every author's famousWorks array and preserves the title, publication year, author name and author ID. The page presents the complete six-book collection as a JSON response at the required GetAllBookAPI route.

```

GetAllBookAPI response
src/views/GetAllBookAPI.vue

1  <script setup lang="ts">
2  import { computed } from 'vue'
3  import authors from '../assets/json/authors.json'
4
5  const apiResponse = computed(() => ({
6    success: true,
7    count: authors.reduce((total, author) => total + author.famousWorks.length, 0),
8    books: authors.flatMap((author) =>
9      author.famousWorks.map((book) => ({
10        title: book.title,
11        year: book.year,
12        author: author.name,
13        authorId: author.id,
14      })),
15    ),
16  }))

```

Figure 8. GetAllBookAPI transformation and JSON response structure.

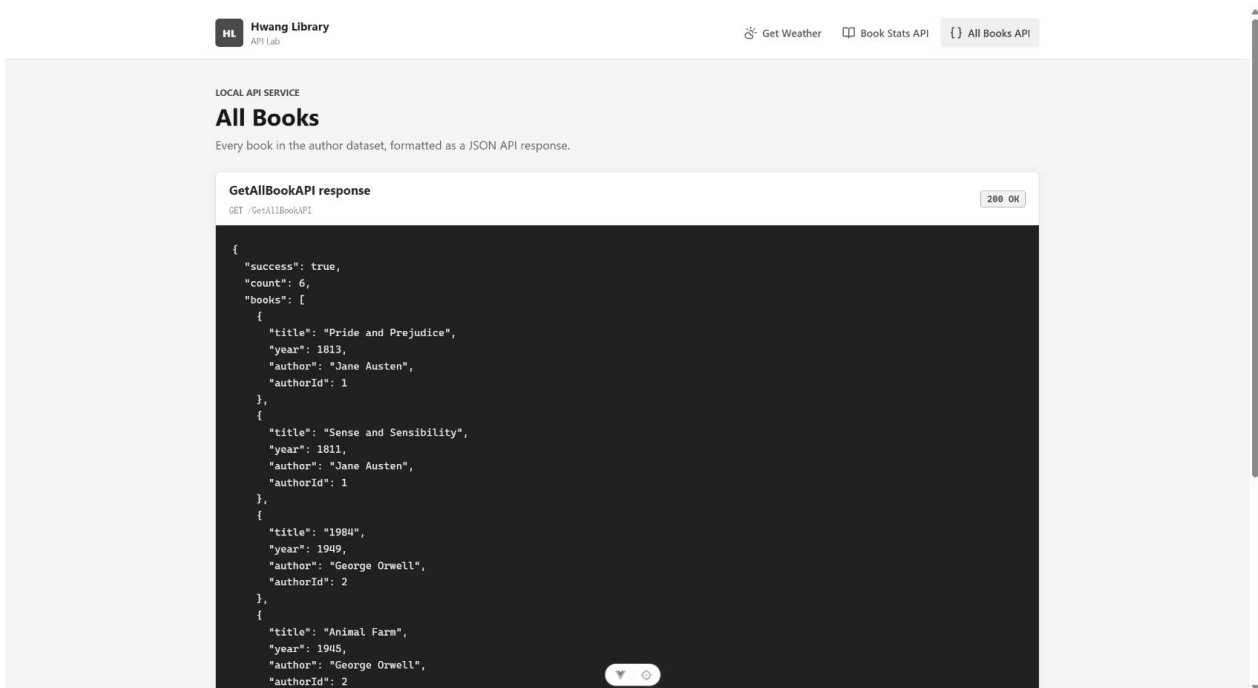


Figure 9. GetAllBookAPI browser response showing count 6 and the first response records.

```
{
  "title": "Pride and Prejudice",
  "year": 1813,
  "author": "Jane Austen",
  "authorId": 1
},
{
  "title": "Sense and Sensibility",
  "year": 1811,
  "author": "Jane Austen",
  "authorId": 1
},
{
  "title": "1984",
  "year": 1949,
  "author": "George Orwell",
  "authorId": 2
},
{
  "title": "Animal Farm",
  "year": 1945,
  "author": "George Orwell",
  "authorId": 2
},
{
  "title": "Murder on the Orient Express",
  "year": 1934,
  "author": "Agatha Christie",
  "authorId": 3
},
{
  "title": "Death on the Nile",
  "year": 1937,
  "author": "Agatha Christie",
  "authorId": 3
}
}
```

Figure 10. Continued browser response showing all remaining books and the closing JSON structure.

6. Verification and Responsive Layout

The project passes Vue TypeScript checking, the production Vite build, and the Vitest suite. The weather-code unit test verifies clear, rain, snow and thunderstorm mappings. Browser checks also confirmed that all navigation labels, controls and results remain readable at a 390-pixel mobile viewport.

```
Weather-code unit test
src/ __tests__ /weather.spec.ts

1  import { describe, expect, it } from 'vitest'
2  import { describeWeatherCode } from '../services/weather'
3
4  describe('describeWeatherCode', () => {
5    it('maps clear, rain, snow and storm weather codes', () => {
6      expect(describeWeatherCode(0)).toEqual({ description: 'Clear sky', kind: 'clear' })
7      expect(describeWeatherCode(61).kind).toBe('rain')
8      expect(describeWeatherCode(75).kind).toBe('snow')
9      expect(describeWeatherCode(95).kind).toBe('storm')
10    })
11  })
```

Figure 11. Unit coverage for external weather-code mapping.

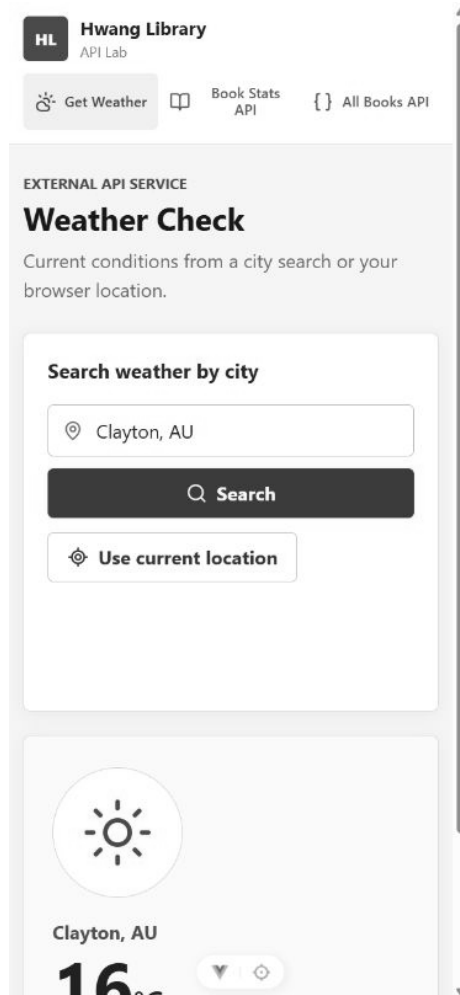


Figure 12. Responsive Weather Check page at a 390-pixel mobile viewport.

7. Git Version Control

Lab 10 was developed on the dedicated lab10-api branch so the existing Lab 2 and Lab 3 history remained intact. The implementation and automated verification were recorded in separate focused commits before the report was added.

```

Git commit history
branch: lab10-api

* cd4f9c8 (HEAD -> main, origin/main, origin/lab10-api, origin/HEAD, lab10-api) Make report headings black
* cb9d367 Add current-location evidence to Lab 10 report
* 06a4967 Add assessed Lab 10 PDF report and evidence
* bf24114 Add Lab 10 tests and setup documentation
* e405141 Implement Lab 10 weather and book APIs
* 1f08632 Delete
* 71bf5ae Complete Lab 3 Bootstrap user information form
* b81093e Set up Bootstrap 5
* 4c9aca6 Create basic user information form
* 56e86c4 Finalize commit
* f6ae68a Add George Orwell highlight binding
* 5e433b2 Add conditional rendering activities
  
```

Figure 13. Git history showing the focused Lab 10 implementation and test commits.

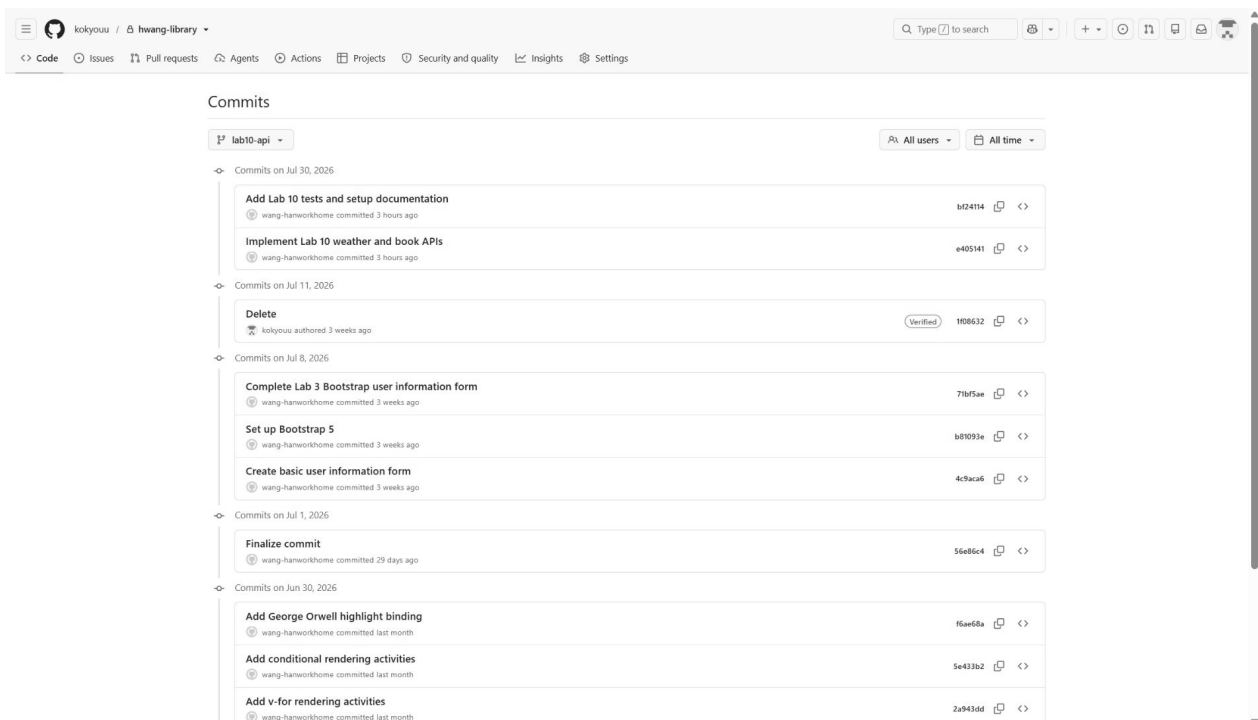


Figure 14. GitHub commit history for the pushed lab10-api branch.

8. Conclusion

The completed Lab 10 application satisfies the Pass/Credit and Distinction/High Distinction functional criteria. It can retrieve weather using browser coordinates or a city query, displays temperatures in Celsius with a matching icon, calculates the author and book totals from the existing dataset, and shows all books as valid JSON. The implementation is tested, responsive, documented and maintained through a clear Git commit history.